

Subtract 00:06:10 from timestamps to find place in audio/video recording.

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00:06:10.080 --> 00:06:15.110

Jamsheed Shorish: Do we have quorum? Should we start, or do you guys want to wait for someone else?

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00:06:16.100 --> 00:06:29.370

Jamsheed Shorish: I think we can probably start. We're at 7 after, we usually try to wait till 5 after. People may trickle in and just jump in as they come. But, yeah, maybe with the time that we have, maybe make sure that we don't lose too much.

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00:06:29.370 --> 00:06:38.500

Jamsheed Shorish: Let's go ahead and do so, and again, Chris, thanks again for, for jumping in with your second of three of your, of your, currently slotted, EconPod,

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00:06:38.520 --> 00:06:46.380

Jamsheed Shorish: let's call them progressions through prediction markets. So I guess today is a bit more on the formal side, or formalisms, or specifications side, that you're going to be looking at?

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00:06:46.880 --> 00:07:09.979

Krzysztof Paruch: Oh, yeah, definitely. So, since the last time we spoke, I think it was roughly two, two months ago, I focused almost entirely on the formal side of things, so I did complete the publication, so I, I deployed the product on publicly, so this is what I did immediately after our conversation last week, last, last time.

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00:07:09.980 --> 00:07:15.100

Krzysztof Paruch: So, so you can actually access the architecture that I have built already,

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00:07:15.100 --> 00:07:20.999

Krzysztof Paruch: on the internet, but I have focused in my work, I have focused mostly on

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00:07:21.670 --> 00:07:44.970

Krzysztof Paruch: on the research stuff that is related to the specification. So, you are familiar with this, right? Like, this is

what we talked about last time, and you see, kind of, like, this thing. It's already online, so that you can access it, and I mean, there is nothing to do, so, like, you probably shouldn't go and do anything, because you cannot do anything yet, but I managed to host it publicly, which is an important step for, kind of, future testing and also deployment.

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00:07:44.990 --> 00:07:50.700

Krzysztof Paruch: But, what I want to show you, and what we will focus on today.

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00:07:50.970 --> 00:07:58.139

Krzysztof Paruch: is, this. So, I have, started writing up, kind of, like, all the information that,

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00:07:58.330 --> 00:08:12.359

Krzysztof Paruch: leads me towards building this, because... and I get... I got a little bit carried away with that title, so I don't know if we will focus so much on signal processing in this publication, but it's kind of like, you know, in honor of our previous work.

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00:08:12.360 --> 00:08:26.939

Krzysztof Paruch: I just wanted to tie the knot here a little bit. So, the write-up here contains, roughly the three sections, or three... three topics that we could discuss, and they are...

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00:08:26.950 --> 00:08:51.749

Krzysztof Paruch: as it is a work in progress, it's, they are independent, obviously, but interdependent, but they are not yet fully coherent. So, you will see when we talk, it's true that there are three distinct things that I want to represent, and I built them up from the bottom up, and this is why there are some moments at which things don't add up yet. So, I guess that the next step would be

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00:08:51.750 --> 00:09:16.720

Krzysztof Paruch: synthesis of those three streams into one coherent framework. But what I did is I started with a GDS spec of a prediction market, just kind of, like, from an engineering standpoint, because I was wondering, okay, if I really want to implement this thing, and I do have the infrastructure already publicly hosted, but I don't have the whole market logic yet, right? So if I want to implement it, I need

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00:09:16.720 --> 00:09:22.809

Krzysztof Paruch: to understand what I'm implementing, what I'm building, which states I have to,

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00:09:22.840 --> 00:09:32.849

Krzysztof Paruch: to consider what are the mechanisms, how do they operate, right? So there is a bottom-up states-based representation of a prediction market, or prediction market types.

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00:09:33.340 --> 00:09:40.329

Krzysztof Paruch: Then you have a discussion of the different market types, which is the order book and AMM.

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00:09:40.330 --> 00:09:56.150

Krzysztof Paruch: class, and how this fits into this GDF framework that I'm proposing here. And finally, in a distinct workstream, I tried to motivate the definition of an AMM

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00:09:56.230 --> 00:10:21.189

Krzysztof Paruch: based prediction market from the scoring rule approach. So, kind of showing that, okay, when we start with scoring rules, and when we have proper scoring rules, we want... we will end up in a setting where we can build a prediction market that is then akin to an AMM, and that demonstrates all the properties from the dynamic system standpoint that we have

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00:10:21.190 --> 00:10:22.560

Krzysztof Paruch: discussed already.

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00:10:22.560 --> 00:10:41.470

Krzysztof Paruch: And then there are some properties associated with them. So this is what I have so far, and it's mostly work that I use to inform the implementation, or the design of the system. So it's kind of like going back and forth between, you know, economic analysis, the design questions, into the implementation side of things.

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00:10:41.470 --> 00:10:46.819

Krzysztof Paruch: But, yeah, for today, I have, fixed or 8 pages.

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00:10:46.830 --> 00:10:49.850

Krzysztof Paruch: That, I can... I can talk you through here.

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00:10:50.030 --> 00:10:51.760

Krzysztof Paruch: If there aren't any questions.

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00:10:55.830 --> 00:11:01.929

Krzysztof Paruch: So what you see here, no abstract, no introduction, and I start with the specification, and I...

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00:11:02.010 --> 00:11:17.399

Krzysztof Paruch: I mean, the event space that I introduce immediately is... I mean, the first thing that I thought when I wanted to represent this as a state space GDS framework is basically the event space. So we have an uncertain event, and the

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00:11:17.400 --> 00:11:40.750

Krzysztof Paruch: potential outcomes of this event. So, like, you have an E and an A, and then you say that these outcomes represent the possible realization of the event E, right? So this would be basically the event side of a prediction market, and then you can add to this the economic side of the prediction market, where you would basically say, okay, there are all the additional things that you need to care about when you really want to see how the system operates.

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00:11:40.750 --> 00:11:58.400

Krzysztof Paruch: Right? So, like, in addition to the event space and the outcomes, you also have an S, and an L, which is basically the time interval during the market is active. You have a vector of outstanding shares corresponding to each outcome.

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00:11:58.400 --> 00:12:03.960

Krzysztof Paruch: So each possibility A_i is represented by a share as I.

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00:12:04.070 --> 00:12:21.800

Krzysztof Paruch: And then you have a total amount of liquidity or collateral backing the outstanding shares. And when I wrote this, I was kind of, like, new to this whole, prediction market, and understanding fully the prediction market thing. So, like, you will see why this is kind of, like, this bottom-up approach, evolving towards something where

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00:12:22.140 --> 00:12:30.849

Krzysztof Paruch: We might see that it doesn't really fit yet, so, like, so why it should, collapse into something that makes sense eventually.

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00:12:31.320 --> 00:12:33.610

Krzysztof Paruch: Okay, so,

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00:12:33.710 --> 00:12:45.089

Krzysztof Paruch: After this is introduced, I introduce, from the work that we've done before, kind of like the state-space representation. So we have this notation of, feasible actions.

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00:12:45.090 --> 00:13:02.389

Krzysztof Paruch: And then the question about how those actions mutate those system states that we have introduced here. And I start with different market mechanisms that are under consideration. So, first of all, the initialization. So, like, how do you set up a prediction market? How do you start?

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00:13:02.530 --> 00:13:18.900

Krzysztof Paruch: And so... so you basically define all of those things that you need. So, saying that we have to specify the event, we have to define the possible outcomes, those outcomes have to be exhaustive and exclusive, so that's... so those are the properties of those.

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00:13:18.900 --> 00:13:23.380

Krzysztof Paruch: And then you define the interval during which the market is active.

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00:13:23.730 --> 00:13:25.359

Krzysztof Paruch: And then,

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00:13:25.360 --> 00:13:48.219

Krzysztof Paruch: yeah, you can say that the T is defined implicitly, obviously, to the formulation of the questions, but sometimes not. So, like, if you say, okay, will something happen before January 1st, you kind of, like, implicitly define already what the time interval is that you're looking for... looking at, but not really, because you can make the market be active or, like, inactive for a period even preceding this date, right?

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00:13:48.520 --> 00:13:50.520

Krzysztof Paruch: Okay, so, just a question.

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00:13:50.520 --> 00:14:07.889

Jamsheed Shorish: Yep. Sorry, just wanted to get a quick clarifying question, just with regard to that specification of how you define the event. If you scroll up and scroll back to that, like, event that's been dictated by time, you know, will this occur before January 1st, for example? That in itself, then, is an outcome

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00:14:08.780 --> 00:14:14.729

Jamsheed Shorish: Should we be looking at that particular type of statement as something as being one of the AIs?

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00:14:15.100 --> 00:14:30.709

Jamsheed Shorish: In other words, is that actually an outcome of the event itself, or is the event itself something else that occurs that itself is then something which then has another inheritance of a random variable whose outcome is coming before or after January 1st? How should we be interpreting that?

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00:14:30.930 --> 00:14:39.489

Krzysztof Paruch: Yeah, so the way how I was thinking about this was kind of like, the event is the question, so will this happen, yes or no? And the A's would be then a yes or a no.

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00:14:39.700 --> 00:14:41.270

Krzysztof Paruch: So those would be the outcomes.

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00:14:41.740 --> 00:14:48.880

Krzysztof Paruch: And this is how I was looking at this. But to your point, I guess, that if you think about this more carefully.

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00:14:49.440 --> 00:15:04.410

Krzysztof Paruch: you might want to, you might end up, collapsing this somehow into something that is more, implicitly defined already through the event, so not really, so assuming that E is the random variable, and then, and then seeing how this,

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00:15:04.570 --> 00:15:06.420

Krzysztof Paruch: How this can be framed differently, right?

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00:15:06.420 --> 00:15:30.689

Zargham: The other question, and maybe this is sufficient as long as your event model is rich enough, most prediction markets in practice need to have a, sort of, like, non-resolution resolution, so, like, they have an outcome that is in favor of, like, yes and no, each that... and, like, the logic for what would happen, but because there are...

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00:15:31.080 --> 00:15:46.989

Zargham: you know, it's possible to have, like, an outcome that is not clearly either. There's usually a sort of dissolution or unwind. Again, you don't have to know a priori exactly what that will be, but there's a...

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00:15:47.050 --> 00:16:11.530

Zargham: I... again, I'm talking more from the place of the production systems. Like, if you look at them carefully, they have, they have conditions and rules for the dissolution and unwind of a collection of positions for edge case handling. Obviously, you need a way of formally defining that boundary so that you can resolve to yes, no, or the... that... that kind of...

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00:16:11.620 --> 00:16:28.929

Zargham: degenerate state, but, like, the ability to have a state in your event model for resolved to not a resolution, and then to have a rule for how the positions are unwound as a function of that is actually quite important.

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00:16:31.330 --> 00:16:35.450

Krzysztof Paruch: Right. So what does this mean, practically, for this definition?

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00:16:35.450 --> 00:17:00.000

Zargham: that you should have a... you should have a tripartite logic. You should have a... it happened, it didn't happen, or, like, basically an edge case handler bit, so that when you write your economic logic for what happens in order to compute expectations and whatnot, you actually have three outcomes. There's the, it happened before, it didn't happen before.

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00:17:00.000 --> 00:17:12.900

Zargham: Or you got the boundary condition case, which is the indeterminate one, and you have a rule set for what happens in the indeterminate one, in addition to the rule sets for what happens for each of the yes and the no.

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00:17:13.910 --> 00:17:28.959

Zargham: Like, it's a model where there are three kinds of outcomes instead of two, where it's clearly yes, clearly no, and... or like, you know, the oracle says yes, the oracle says no, or you could think of it as the oracle breaks.

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00:17:28.960 --> 00:17:29.560

Krzysztof Paruch: Oh, yeah.

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00:17:29.560 --> 00:17:40.250

Zargham: breaks, what happens? And the answer is usually something like a predetermined unwinding policy that is different from the yes wins and no wins.

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00:17:40.890 --> 00:17:44.940

Jamsheed Shorish: There's a... there's an alternative that you might also entertain that,

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00:17:45.240 --> 00:17:57.920

Jamsheed Shorish: in essence, kind of folds in this edge case into the ability of individuals to distinguish between events, which is to use a sigma algebra over the possible outcomes, and say, well.

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00:17:57.930 --> 00:18:16.969

Jamsheed Shorish: You can't observe something with certainty, necessarily. You can only observe things that occur within a particular subset of possible outcomes, which may be singleton sets, in which case you get the, I got it, it did happen, or it didn't happen, but you could also have a larger set of outcomes that you... that your information can't refine.

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00:18:17.180 --> 00:18:35.379

Jamsheed Shorish: Because the idea of that tripartite, that third outcome as being, hey, this is the edge case where we can't resolve, can be specified in the sigma algebra approach as saying, well, then what you're doing is you're saying it's resolvable up to a subset of outcomes, since you know that something has to happen, or has to have happened.

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00:18:35.440 --> 00:18:41.409

Jamsheed Shorish: So that would leverage a lot of the random variable,

approach, where basically you would be treating the

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00:18:41.410 --> 00:18:59.250

Jamsheed Shorish: the entire thing is as a kind of a probability space. You would have a random variable, you would have its particular outcome set, then you would have the sigma algebra, the set of subsets of all possible outcomes that then become the information partition for the people who are submitting their probability distributions over those outcomes.

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00:18:59.470 --> 00:19:14.729

Zargham: Yeah, that makes sense to me. I would just note that, like, these are, like, coming from opposite directions. One is how you would enrich the formalisms to most properly address this, like, fundamental gap in certainty over whether certain things happened.

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00:19:14.730 --> 00:19:22.100

Zargham: And the other is coming from the direction of, like, just how prediction markets are implemented and how this

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00:19:22.100 --> 00:19:26.580

Zargham: Fundamental problem is... has been resolved in practice to date.

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00:19:27.510 --> 00:19:52.469

Krzysztof Paruch: Right, good point. So, like, first of all, to the point of, kind of, you know, a tiebreakers, like, I haven't really considered this at all. So, to me, it was all events are exclusive, exhaustive, and you can... yeah, you cannot really have anything in between, like, kind of looking at this scenario, but I... I take your point here, that it's a valid argument to make. And with regards to the continuity here, this is a very interesting

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00:19:52.470 --> 00:20:08.959

Krzysztof Paruch: interesting consideration. I've been approaching this topic from multiple directions. Currently, as you see, kind of like a discrete case, but I also have, on my, you know, research, a few, like, expanding this into a continuous approach. So here, the definition of a probability space.

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00:20:08.960 --> 00:20:20.519

Krzysztof Paruch: And the measuretical approach is definitely something worth considering. And this is basically where you end up

when you, you know, like, start here and then see how far things go.

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00:20:20.650 --> 00:20:32.340

Krzysztof Paruch: And there is a lot of literature already that I've found that, you know, things of those spaces in terms of continuous spaces. So, yeah, definitely, we will... we will see those things eventually coming up.

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00:20:32.340 --> 00:20:42.639

Jamsheed Shorish: Yeah, just to clarify, you don't need continuity. You can still have a discrete event space. It's just that you have uncertainty over what's happening. So, for example, if you are

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00:20:42.640 --> 00:20:58.909

Jamsheed Shorish: rolling a 6-sided die, and for whatever reason, you can only distinguish between the sets. A 1 or a 2 is rolled, a 3 or a 4 is rolled, or a 5 or a 6 is rolled, but you can't distinguish between 1 and 2, 3 and 4, or 5 and 6.

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00:20:58.910 --> 00:21:08.560

Jamsheed Shorish: then that's essentially giving you the set of subsets that you can yourself distinguish, and it's an information partition of the entire sigma algebra.

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00:21:08.560 --> 00:21:22.389

Jamsheed Shorish: over the possible combinations of things that all could happen in terms of the ways in which you might not be able to distinguish, for example, between 1 through 5, and you can only distinguish 6 when it occurs, for example. That's one particular information partition.

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00:21:22.390 --> 00:21:32.589

Zargham: ways you characterize observability, and, like, ultimately, in different traditions, you'll get... you'll get different focuses. Again, the... the probability theory stuff is...

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00:21:32.590 --> 00:21:44.580

Zargham: kind of baked into some of the stochastic control stuff, or the statistical signal processing fields, which you seem to be drawing on here. But to kind of tie into what Jamshid's saying, I think

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00:21:44.580 --> 00:21:52.959

Zargham: Fundamentally, you can't assume perfect observability here, and it just... in both in theory and in practice, it is...

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00:21:52.960 --> 00:22:07.639

Zargham: the case, that there will be limits on your ability to distinguish things. And so you have to make some decisions, I guess, at this paper level, whether you're gonna completely abstract that away, as you have initially here, versus, like, what level of depth you want to take that on.

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00:22:07.640 --> 00:22:20.259

Zargham: In my opinion, you have 3 broad bucket choices. One is ignore it and consider a theory where you just know, which is sort of the way it's written now. One is the other extreme, which is to go all the way to

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00:22:20.260 --> 00:22:44.480

Zargham: kind of the measure theoretic probability sigma algebra stuff, and, like, really engage with observability in a full-throated way. And the other is to, apply something in the middle where you make some assumptions about the limits of your observability, say that there's, you know, cases in which the decision is not strictly observable, that lands you in a degenerate case, and you make sure that you have a

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00:22:44.480 --> 00:23:02.420

Zargham: a process for handling that degenerate case, but now observability isn't the thing that you're writing your paper about. You're just having a third case in your logic to account for the fact that incomplete observability is a fact of life and should be accounted for in the design of prediction markets.

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00:23:03.240 --> 00:23:06.979

Krzysztof Paruch: Right. Good point. No, I'll take this, up. Thank you.

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00:23:06.980 --> 00:23:23.980

Krzysztof Paruch: Yeah, so based on, kind of, like, the framework that we've introduced here, I'll try to push a little bit, because we have so much ground to cover, so I really want to move forward. But, based on the initial framework developed, I look into different mechanisms. So the market initialization mechanism, for example, is just kind of, like, saying, okay.

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00:23:23.980 --> 00:23:46.790

Krzysztof Paruch: What is the initial market stage? Like, how many, how many shares do we have, for example, initially? Like, what is the parametrization of the curve, when we're looking at an AMM-style prediction market, or what is the, what is the inventory of the order book, for example, or the supply of the outfilling shares?

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00:23:46.790 --> 00:23:48.509

Krzysztof Paruch: These kind of things, right?

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00:23:48.710 --> 00:24:02.439

Krzysztof Paruch: No, sorry, sorry, I'm mixing this up. So the marketization is actually just the definition of the question, I guess, and this is how I defined it. So, like, you basically just say, what is the event, what is the question, what are the outcomes, and what is the time?

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00:24:02.440 --> 00:24:09.059

Krzysztof Paruch: duration. We move into those questions when we talk about the primary and secondary market structure.

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00:24:09.060 --> 00:24:25.820

Krzysztof Paruch: although leveraging the framework introduced, we are saying here that, okay, you have a primary market or a secondary market, and one of the things that I didn't know back then very well then when I wrote this, but I...

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00:24:25.820 --> 00:24:38.120

Krzysztof Paruch: in the meantime know, is that in the order book case, you have a clear distinction between the primary and the secondary market, and in the AM case, where you...

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00:24:38.120 --> 00:24:50.529

Krzysztof Paruch: traded on a curve that is derived from a scoring rule, those markets collapse into one. So, like, you do have situations where the primary and the secondary market are the same, so you don't really make a clear distinction between them.

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00:24:50.830 --> 00:25:15.109

Krzysztof Paruch: And this is starting from the current operations of this primary market. So you can see what are the eligible mechanisms, what is it that we can do, is we can add liquidity, which means that

we are basically minting a full set of outcome shares. So we are saying, okay, for one unit of, of payout, so one

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00:25:15.460 --> 00:25:22.830

Krzysztof Paruch: \$1, for example, that I... that I pay, I can mint one of each shares, because at the end.

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00:25:22.830 --> 00:25:45.180

Krzysztof Paruch: we will have only one of those shares winning, so this is the one that will be eligible for the one unit of payout at resolution. So you can do this at any time, so, like, you don't change anything, and this is the liquidity addition mechanism, or the share minting mechanism on this primary market that basically everyone can do at any time.

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00:25:45.630 --> 00:26:06.109

Krzysztof Paruch: And, here I formalized, basically, those, those claims, so I'm saying that each share, like, the winning share is entitled to claiming P -bar at resolution, and that this all works out, because, yeah, there is basically just, just only one share that wins, and others will be forfeited at settlement.

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00:26:06.820 --> 00:26:24.439

Krzysztof Paruch: So... so this is then tracking, from a... from this DDS perspective, this is tracking the evolution of the market. So you're basically saying, okay, the event and the answers stay the same, the times stay the same, but you are basically adding just, an amount ΔS , the amount of liquidity, P bar that you've added.

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00:26:24.440 --> 00:26:34.239

Krzysztof Paruch: No, sorry, like, here you are minting this many full sets of shares, so ΔS , for an amount ΔL , which is the amount of liquidity that you've added to this pool.

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00:26:34.700 --> 00:26:44.239

Krzysztof Paruch: And this works out, this is very nice. And here, also, you can look at it from the agent perspective, where you basically say, okay, you just pay something into the pool, and you receive the ΔS .

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00:26:44.580 --> 00:26:53.010

Krzysztof Paruch: Okay. On the secondary market, however, we allow agents to trade, between those shares, and now.

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00:26:53.170 --> 00:27:16.489

Krzysztof Paruch: we just enable the trade mechanism. So we say, okay, what you can do is you can, at any time, you can say, I want to change some amount of a share that I hold, of my shareholding, with respect to another share, another outcome or another asset that I want to obtain, and I just want to say, okay, I can place a bet

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00:27:16.490 --> 00:27:21.560

Krzysztof Paruch: Or place a bid, actually, on the order book, and say.

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00:27:21.560 --> 00:27:37.900

Krzysztof Paruch: okay, this many Delta SE shares I want to exchange against this much, and then against this one asset, and then the market will dictate, like, that those prices here will arise from the market activity. So, so there is no...

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00:27:37.900 --> 00:28:02.650

Krzysztof Paruch: price definition here, or no price derivation, it's just, you know, coming from the order book. And it works out pretty well, and you see that those trades don't affect the liquidity, they don't affect outstanding shares, it's a zero-sum game, but there is kind of, like, this price that is then adjusting that you have to look at and keep track separately, somehow. So the question would be.

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00:28:03.160 --> 00:28:09.739

Krzysztof Paruch: how do we treat the price now? And I mean, you can look at it from the perspective of a metric, right?

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00:28:10.110 --> 00:28:23.880

Krzysztof Paruch: That is, that is derived from, from the state that you have. And then, market resolution, finally, which is, when you know, what the realization of the, of the, event is.

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00:28:23.960 --> 00:28:33.490

Krzysztof Paruch: then you can say, okay, this is the one that wins, and, kind of each share qualifies for the payout, P -bar, and receives nothing otherwise.

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00:28:33.700 --> 00:28:36.089

Krzysztof Paruch: So, so then you can make this claim.

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00:28:36.800 --> 00:28:43.930

Krzysztof Paruch: And, Yeah, what do I have here, and base implementations,

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00:28:47.370 --> 00:29:08.809

Krzysztof Paruch: yeah, basically just describing what I said. So, so this is... and again, so when I, when I wrote this bottom-up, I was trying to encompass the full prediction market setting, always thinking a little bit about, kind of, like, the order book scenario, but knowing that, okay, there is an AMM logic to it as well, from a design perspective, so, how does this fit, and how does this work together? So...

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00:29:08.810 --> 00:29:19.490

Krzysztof Paruch: There is definitely more to be said about, kind of, like, the primary and secondary market when thinking about the AMM setting compared to the order book setting.

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00:29:20.140 --> 00:29:43.370

Krzysztof Paruch: So, this is what I do here. So, like, I introduce the market types, and I say, okay, like, those are the different types, and again, from a narrative perspective, it might not be the best way to talk about those things in this format. So, like, maybe, like, once I have everything defined and formulated, I will decide to rearrange some of the sections and just, like, try to introduce a different narrative flow, because

149

00:29:44.250 --> 00:29:59.000

Krzysztof Paruch: it made sense when I wrote it, but when you try to kind of, like, you know, explain all the special cases, maybe something more anchored from the price, or more anchored from scoring rules would be the natural way to talk about those things.

150

00:29:59.640 --> 00:30:09.780

Jamsheed Shorish: One, one, one quick question, just as another clarification on the GDS approach that you'd adopted. Is there a reason why the...

151

00:30:10.610 --> 00:30:14.950

Jamsheed Shorish: Random variable. The event is resolved.

152

00:30:15.140 --> 00:30:30.120

Jamsheed Shorish: isn't brought in in a GDS way as an external environmental factor that comes in and helps determine the state. That way you can say it depends upon the realization, and that way you don't have to have...

153

00:30:30.200 --> 00:30:39.009

Jamsheed Shorish: multiple types of state transitions. Instead, you have whether or not this particular random variable has been realized.

154

00:30:39.150 --> 00:30:46.130

Jamsheed Shorish: as dictating the input into the GDS from the outside, from the Oracle universe.

155

00:30:50.350 --> 00:30:55.929

Krzysztof Paruch: So, what is it that you propose? Saying that E is.

156

00:30:56.770 --> 00:31:02.489

Jamsheed Shorish: So you've got the... you've got the event itself, which is E, which has made the state

157

00:31:02.930 --> 00:31:07.770

Jamsheed Shorish: So if you go back up one to just the regular definition of the state, Yeah.

158

00:31:08.330 --> 00:31:09.700

Krzysztof Paruch: Which would be here, right? Yeah.

159

00:31:09.900 --> 00:31:14.889

Jamsheed Shorish: Yeah, exactly. So, the... the tuple... E and A.

160

00:31:15.200 --> 00:31:31.840

Jamsheed Shorish: is that to be interpreted as, like, I identify the kind of event that I want to bet on, and here are its possible outcomes? Or is that interpreted as a random variable that is currently uncertain that has a particular outcome realization? Because

161

00:31:31.870 --> 00:31:47.529

Jamsheed Shorish: It's the latter one that ends up driving payoffs. It's the realization of the random variable, and so it would be more natural to introduce that as something that comes in as an

environmental effect. It's a thing that you can't see

162

00:31:47.530 --> 00:32:01.130

Jamsheed Shorish: But you can make claims on and bets on potential outcomes of, but it doesn't actually occupy the state of the system itself. Otherwise, there's a danger of the interpretation of the state space as being random itself.

163

00:32:01.350 --> 00:32:02.329

Jamsheed Shorish: Which you don't want.

164

00:32:02.930 --> 00:32:14.519

Krzysztof Paruch: Right, okay, I see your point now. So, the reason why it's stated like this is because, again, bottom-up, I was coming from the implementation side, like, how do I build this, and, like, what do I need to code? What do I need to code? Which states do I need to track?

165

00:32:14.570 --> 00:32:38.879

Krzysztof Paruch: And it sounded, like, logical to me, the first thing that I want to do is, like, to set up a market, like, I need to define those things, and this is the protection that I need to do, right, without even worrying about anything. So this is where you see, kind of, like, this perfect section here, being grounded. But it's definitely, as you say, not... maybe not the cleanest approach here. So, what you will see in the latter section, then, when we talk about, scoring rules.

166

00:32:38.880 --> 00:32:56.069

Krzysztof Paruch: you get, kind of, like, this probabilistic, description. So you say, this is a random variable, and you can make claims about, kind of, like, the realization of this variable, and then, based on, kind of, like, the difference between your claim and the realization, you get some score out of it, right? So,

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00:32:56.070 --> 00:32:56.840

Krzysztof Paruch: So this is...

168

00:32:56.840 --> 00:33:13.840

Jamsheed Shorish: Which makes perfect sense, yeah. The only question then is just to make sure that when you go back, if you want to retain the GDS specification of that process, that it then has to be elevated into sort of a fully formed random variable, and then brought in

something as something coming in from the outside. Then those two approaches are entirely consistent.

169

00:33:14.240 --> 00:33:36.470

Krzysztof Paruch: Right, and this is the main goal, so, like, actually, I believe that I want to show that, demonstrate that you can actually define a framework that encompasses different market designs, or, like, different market formulations, and that it is consistent between, kind of, like, the... to write it as a stochastic variable and a state-based representation, and, like, this is what I'm trying to...

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00:33:36.670 --> 00:33:37.420

Zargham: So, I, I think...

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00:33:37.420 --> 00:33:37.910

Jamsheed Shorish: Fuck.

172

00:33:37.910 --> 00:33:38.270

Zargham: to...

173

00:33:38.270 --> 00:33:38.780

Jamsheed Shorish: Yeah.

174

00:33:38.780 --> 00:34:03.530

Zargham: Oh, sorry, I'll make my point and then let go. Like, I think in terms of tying it back to GDS, the thing that's coming to me is still this discussion about the sigma algebras and the observability, because our earlier GDS work was so attentive to, kind of, reachability, which is related to the controllability side of a dynamical systems problem, and that, like, what you're doing when you move into this questions of signal processing.

175

00:34:03.530 --> 00:34:28.479

Zargham: and of, in particular, prediction markets is you're poking at the part of the problem that I think that we left comparatively underspecified, because you could say that it's possible in a dynamical systems representation that a particular state is reachable, but, like, you actually don't know whether you can know that you reached it. And so there's a, at least for me, like, as I'm thinking through what you're sharing and thinking of how I would map it onto GDS formally.

176

00:34:28.480 --> 00:34:37.480

Zargham: It does invite discussion of distinguishing, like, re... the ability to reach states from the ability to know that you've reached states.

177

00:34:38.409 --> 00:34:42.129

Jamsheed Shorish: Well, I think that's fine, and I think that insofar as you have a...

178

00:34:42.190 --> 00:35:05.459

Jamsheed Shorish: you know, we were, as you said, we were quite agnostic as to how the input space and the decision space is dependent upon the state space, and so we could just as easily have had that scriptU as incorporating a probability distribution over a subset of states, and you're like, well, I don't know which one I'm actually at, but I'm probably in one of these, and that way you would be able to kind of incorporate that, I think, seamlessly, but the

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00:35:05.460 --> 00:35:08.449

Jamsheed Shorish: That's introducing, kind of, an assessment of the

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00:35:08.450 --> 00:35:16.550

Jamsheed Shorish: the sensor apparatus, let's call it, of the, of the user. And my question's a little more fundamental, just to... for Chris to...

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00:35:16.590 --> 00:35:34.459

Jamsheed Shorish: ensure that a random variable that is, in fact, random to the entire system is being treated as something that is not actually considered to be something with knowledge about, as in a state, because it looks strange, in a sense, to encode a random variable as the state.

182

00:35:37.410 --> 00:35:50.599

Krzysztof Paruch: Yeah, yeah, so this is a good point, yeah. I would think about this, or, I mean, put it on my to-do list when I'm reconciling those approaches, and as you said, making sure that it is consistent throughout.

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00:35:50.600 --> 00:36:15.590

Zargham: That makes sense. I just... I want to keep pushing on the, like, the ability to make claims about whether something is processing signal does necessarily relate to the potentially very diverse

measurement apparatus. So you have a layered system where you have some amount of state, some of which might not be observable to anyone. You have a bunch of actors who have their own measurements that have their own processes

184

00:36:15.590 --> 00:36:40.070

Zargham: through which they ascertain some partial, incomplete, potentially local information to themselves, and then you have an aggregate level process that's structured by a market, right? So we can assert some rules about how part of the state evolves, and then try to drive that system, or excuse me, have that system be driven by the behavior of the actors who themselves have their own local measurement apparatus.

185

00:36:40.070 --> 00:36:57.570

Zargham: And in order to make claims about this system as a whole, processing or learning information, that can't, by definition, be in the state of the implemented technological system. It's the other way around. It's that the hidden information is buried in some...

186

00:36:57.570 --> 00:37:22.519

Zargham: Something in the measurement or interpretation of the state by the diverse actors with their, like, unique measurement apparatus, and then the ways in which those many unique measurements with their unique measurement apparatus are through the dynamical system, through the market structure, processed into something that we can interpret as being some kind of measure of central tendency.

187

00:37:22.520 --> 00:37:29.190

Zargham: price, measurement, or otherwise. And I'll send you a paper that I like that treats this, for...

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00:37:29.190 --> 00:37:52.889

Zargham: distributed sense-making where a bunch of nodes each have their own individual observations of a private process, and then they naively share their beliefs, not their measurements, and then there are proofs about the extent to which an apparatus where no one actor has enough information to discern the underlying, you know, true state of the world.

189

00:37:52.890 --> 00:38:16.260

Zargham: But that the naive belief-sharing process combined with certain criteria over the local measurement processes still gives you convergence to the true state of the world. And at least, and this is,

again, my opinion, but, like, if you're poking at a signal processing interpretation and trying to kind of reconcile prediction markets as filters, as tools for ascertaining hidden information, then that's

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00:38:16.260 --> 00:38:23.569

Zargham: probably the part of the literature to poke around. And again, I'm not saying you have to force it into this paper, I'm just, like.

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00:38:23.650 --> 00:38:46.779

Zargham: I think that there's a gap in our prior work on GDS, where we... our attention was primarily to reachability, and that actually any closed-loop dynamical system is going to have considerations on observability, which will also affect the more plausible path. So regions of the state space that are, for example.

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00:38:46.850 --> 00:39:10.700

Zargham: harder to make sense of might exhibit, for example, higher variance behavior, or... I don't know, there's a lot we don't know. I'm not saying we have the answer, I'm just saying the biggest thing that I took away from this was a kind of poke at a place where I think there's a lot more thinking and formalisms that could be developed to understand the quote-unquote other half of the generalized control systems problem.

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00:39:15.070 --> 00:39:32.929

Krzysztof Paruch: Nice, no, perfectly, perfectly makes sense. But I agree with you that this might not be the focus of this paper here, so... but it's pointing in the direction that could be explored once this framework is established, I would say, so you could leverage, then, this to... to make about the stuff that you mentioned.

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00:39:32.940 --> 00:39:52.820

Krzysztof Paruch: Okay, so let me go back to what we have here. So we were here at the market types, or the books markets, we talked about this, we defined trading functions. Yeah, I just... maybe this is not so useful, but I did this just to, kind of, like, define, with an eye on, kind of, implementations, like, what do I need to implement, what do I need to do?

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00:39:52.910 --> 00:40:16.649

Krzysztof Paruch: Okay, and then let's talk about the automated market makers, because this is a bit orthogonal to this, which is basically a situation where you replace the peer matching process with an

algorithmic pricing rule that is derived from a cost function, and kind of the whole exercise of showing that this is actually working is what is the remainder of the paper, or is

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00:40:16.650 --> 00:40:18.459

Krzysztof Paruch: Yet to be finished.

197

00:40:18.580 --> 00:40:33.909

Krzysztof Paruch: So, what do we have here? We... we... yeah, they... basically that you trade, along this curve, and that, you can, you can do this. And, and the, the question then is,

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00:40:34.280 --> 00:40:40.850

Krzysztof Paruch: the question then becomes, because in the order book, Ty,

199

00:40:41.480 --> 00:40:57.229

Krzysztof Paruch: case, it's very straightforward that you have solvent, right? So, you... it's obvious, because it's a zero-sum game. You trade between, shareholders, and the prices adjust accordingly. But, the question here is, in the AMM type.

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00:40:57.230 --> 00:41:15.559

Krzysztof Paruch: Just to make sure that you actually can, like, that the prediction market is sold well, so, like, that it really can pay out that resolution in any case, and that it has bounded loss. For the initiator, or the initializer of the protocol, this is something that is worth considering here.

201

00:41:16.500 --> 00:41:39.770

Krzysztof Paruch: Okay. So once this is established, I've moved forward with defining the prices. So, like, there is a section of saying, okay, like, what are the kind of, like, the... we haven't had prices yet, right? Like, we have to introduce them somehow. And then we say that each, each share is priced at a given, PI, according to the, like, the SI, the same index.

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00:41:39.940 --> 00:41:43.369

Krzysztof Paruch: And then... and then, you can...

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00:41:43.630 --> 00:41:48.130

Krzysztof Paruch: you can, you can do this. What do I have here?

Shareholdings recorded public payment?

204

00:41:48.150 --> 00:41:58.450

Krzysztof Paruch: Yeah, exactly. So, like, on the AMM, case, you do have, kind of, like, the, the cost function that is heavily involved, and then you, although

205

00:41:58.450 --> 00:42:09.799

Krzysztof Paruch: are being asked to deposit at Delta L. So, like, you have to... either you receive something from the... from the, from the market, or you have to, to...

206

00:42:09.800 --> 00:42:28.490

Krzysztof Paruch: post something to the market, so deposit something, but this is calculated based on, kind of, like, the difference between the cost function evaluated at the previous shareholding and the current shareholding, or the shareholding that has been obtained by your change of delta S.

207

00:42:29.100 --> 00:42:30.730

Jamsheed Shorish: And then.

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00:42:30.730 --> 00:42:31.250

Krzysztof Paruch: Yep.

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00:42:31.250 --> 00:42:33.419

Jamsheed Shorish: Sorry, just quickly ask, is this the direct

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00:42:33.700 --> 00:42:40.240

Jamsheed Shorish: application of Hansen's approach, then? This is what you're advocating for this particular form, where you have this kind of, like.

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00:42:40.570 --> 00:42:44.879

Jamsheed Shorish: Rebate system in place with regard to the previous values, given

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00:42:45.290 --> 00:42:47.479

Jamsheed Shorish: Or is this somewhere somewhat different?

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00:42:47.480 --> 00:42:58.469

Krzysztof Paruch: Yeah, so I guess so. So the answer to your question here is, I guess, yes. I have a deeper analysis of the hands-on approach here.

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00:42:58.470 --> 00:43:07.709

Krzysztof Paruch: So this would be basically this section, that shows exactly how this is supposed to work, and that you can do this. Okay. Yeah, got it.

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00:43:08.080 --> 00:43:32.539

Krzysztof Paruch: But I do not have the full confidence yet that this is exactly what I need. So, like, I would be happy, like, from an implementation standpoint, that, okay, yeah, I can go with this, because, like, I know exactly what I need to charge the person who wants to make a share adjustment, right? But since I'm still working on it, I have not fully obtained confidence that this is exactly what I need, and I'm still investigating this a little bit, but,

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00:43:32.540 --> 00:43:36.720

Krzysztof Paruch: But once I'm fully confident, then this will be exactly, the...

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00:43:36.850 --> 00:43:40.130

Krzysztof Paruch: Not the function that is being implemented in the mechanism then, yeah.

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00:43:42.110 --> 00:44:00.039

Krzysztof Paruch: Exactly. So, so now we talk about, liquidity endowment and market bootstrapping. So, the next question here with, cost function AMMs is that you don't really need to set up initial liquidity. So, contrary to,

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00:44:00.350 --> 00:44:19.709

Krzysztof Paruch: to the order bookcase, where there is someone who has to be the liquidity provider, who has to buy shares first, and then trades those shares on a secondary market. Here, those two things happen at the same time, and you don't really need to do this, and you definitely don't need to do this in

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00:44:19.710 --> 00:44:23.210

Krzysztof Paruch: And, Market

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00:44:23.310 --> 00:44:37.729

Krzysztof Paruch: scoring rule case. So, because, you can, you can derive those, cost function AMMs, from different, properties, and, once you're,

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00:44:37.730 --> 00:45:02.569

Krzysztof Paruch: initial assumptions are strong enough, then you can say, okay, perfect, those two markets collapse into each other, and you just simply can start trading on the curve without anyone ever having to deposit liquidity. You can improve the trading algorithm by depositing liquidity. So you can say, if there is a liquidity provider or someone who is initializing this, then, you know, effects that we've investigated in the past on AMMs, like

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00:45:02.570 --> 00:45:16.380

Krzysztof Paruch: slippage, will be, will be lower, right? Like, you will have a lower change in, lower price impact on your trades. But... but it's not necessary, inherently, that you have it.

224

00:45:17.850 --> 00:45:36.260

Krzysztof Paruch: So, so this is what, what we have here, and then you can basically do it, but you don't need to do it. Then, the definition of prices, which is also a straightforward consequence of, of what we've talked in the past. So, like, you have this, partial derivative of the cost function,

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00:45:36.310 --> 00:45:48.149

Krzysztof Paruch: in the direction of one of the shares, which is the instantaneous marginal price of one of the shares. And then you can hold the vector of all of the prices and define it like this.

226

00:45:48.610 --> 00:46:08.369

Krzysztof Paruch: look, nice. And then you have the, the bounded loss property, so that you, you really can say that, okay, it would be great to know, how... if, when we are trading on this curve, and people are paying some prices, and then, kind of, like, one of the shares only, is basically,

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00:46:08.370 --> 00:46:28.159

Krzysztof Paruch: paying out a reward, and people have just, you know, like, adjusted those things randomly over time, and have deposited or withdrawn, or have been awarded some of the Delta L during those steps. The question is if we actually can afford to pay out all of the

winning shares, right?

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00:46:28.220 --> 00:46:43.460

Krzysztof Paruch: And... and this is, to my knowledge, so, like, what I understood from the literature, this is... this is possible. So, like, there are, basically, conditions under which you can say that, that this, condition holds, so... so that we have.

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00:46:43.560 --> 00:46:53.120

Krzysztof Paruch: So that we have the availability to cover all of our outstanding, payment.

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00:46:53.120 --> 00:47:16.169

Krzysztof Paruch: And this is especially the case for complex proper cost functions with bounded loss. So we have the logarithmic market scoring rule, which is one of the things that prominently shows up in Henson's literature. So, like, he basically points towards this, okay, this is a good scoring rule, this is what you want, because it has all of the nice properties, and you can, out of this, construct a nice, cost function AMM.

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00:47:16.170 --> 00:47:20.209

Krzysztof Paruch: That is also, having these predictive properties that you're looking for.

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00:47:20.800 --> 00:47:35.149

Krzysztof Paruch: Okay, and since you have those prices, if you normalize them, then you can make the interpretation about probabilistic properties, right? So you can say that those prices, if you normalize them, are actually representing the market belief about the

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00:47:35.440 --> 00:47:40.439

Krzysztof Paruch: The outcome, or the realization, the likelihood of the realization of this event.

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00:47:41.010 --> 00:47:50.210

Krzysztof Paruch: And, and this is how, kind of, like, I, like, again, bottom-up, introduced into the probability, theory here. But,

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00:47:50.210 --> 00:48:06.249

Krzysztof Paruch: it would be worthwhile starting from the other side

as well, right? So saying, okay, we have kind of a random variable, we are looking at probabilities of this variable, we are letting people, to express their opinions about this, and we will aggregate those signals

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00:48:06.250 --> 00:48:18.769

Krzysztof Paruch: Through some cost or punishment function into a market operation, and then pull those signals to pull this information into one aggregate signal.

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00:48:25.230 --> 00:48:39.489

Krzysztof Paruch: And, yeah, any comments before we jump into, basically, what is a literature review of Hansen's paper, which probably shouldn't also go here, but it should go somewhere else, but this is how I, you know, sequentially,

238

00:48:40.000 --> 00:48:43.440

Krzysztof Paruch: acquired this knowledge, so this is why it's only here.

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00:48:45.170 --> 00:49:00.820

Jamsheed Shorish: Well, I guess the... just one comment that maybe you're already going to be getting to is that, you know, now that you've introduced what are called, you know, state prices, essentially the normalized probability distribution, treating the prices then as if they are reflective of likelihood.

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00:49:00.890 --> 00:49:10.450

Jamsheed Shorish: Then you get into the question of the comparison between the objective probability distribution of outcomes, and then the interpretation of

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00:49:10.710 --> 00:49:26.419

Jamsheed Shorish: how individuals value things from a risk-neutral point of view, and so you start looking at equivalent martingale measures. That's going to introduce a behavioral component. I think we touched upon this way back when, in maybe a couple months before, maybe even in the talk before that.

242

00:49:26.480 --> 00:49:45.719

Jamsheed Shorish: About the question of how you want to wrangle the difference in individuals' assessments of what they're going to get according to whether or not they use a risk-neutral measure or they

use the objective probability measure, because you get different payoffs depending upon which one you use.

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00:49:46.010 --> 00:49:59.769

Jamsheed Shorish: And prediction markets themselves, of course, are sensitive to which one is being used. So, I don't know if that's something you're already considering or has already been done, but I just wanted to point it out once again that that, you know, you have that notion of the...

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00:49:59.920 --> 00:50:18.899

Jamsheed Shorish: of the bookmaker's probability distribution versus the objective probability distribution, for example, for a horse race. And the fact that those are almost never the same means that you have to price carefully in order to ensure that individuals are receiving according to how the probabilities are being shaped by other participants.

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00:50:20.120 --> 00:50:21.850

Jamsheed Shorish: Just wanted to mention that once more.

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00:50:22.400 --> 00:50:32.519

Krzysztof Paruch: Yeah, so this is good, and I mean, I'm getting closer to understanding now what you mean by all that, as we're moving forward and progressing here on those definitions, but

247

00:50:32.520 --> 00:50:46.019

Krzysztof Paruch: yeah, I guess that we will, cross this bridge once we... once we get there. So, I have not thought much about this yet, but, isn't this the whole point of it? That you're kind of, like, on an,

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00:50:46.020 --> 00:50:47.979

Krzysztof Paruch: AMM, curve.

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00:50:48.130 --> 00:50:53.820

Krzysztof Paruch: You don't really have to think about it, because you don't have to initialize the market.

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00:50:53.960 --> 00:51:03.830

Krzysztof Paruch: Up front, so, like, the objective, distribution falls under the table, so to say, so, like, you could ignore it to

some extent.

251

00:51:04.930 --> 00:51:11.379

Jamsheed Shorish: Yeah, yeah, that's... I mean, that's true. It is the case that it becomes relatively,

252

00:51:11.390 --> 00:51:24.700

Jamsheed Shorish: let's say, independent of what the underlying probabilities actually are, but at some point, you do need to recover those, and so it's just to make sure that the system is internally consistent. Your state prices that you've... as you've defined them, the pi's, the pi summis.

253

00:51:24.700 --> 00:51:33.970

Jamsheed Shorish: Those have to have well-defined properties, and of course, that's going to depend upon the underlying measure that's being now adopted, just to use a measure theoretic description, because it's easier.

254

00:51:33.970 --> 00:51:45.449

Jamsheed Shorish: To articulate. That has that measure that has to be something which is changing properly to convert the objective probabilities, in a sense, into their risk-neutral equivalents, so...

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00:51:45.610 --> 00:51:48.739

Jamsheed Shorish: But you're quite right, I mean, you get that independence, yeah.

256

00:51:49.380 --> 00:52:01.140

Krzysztof Paruch: Okay, we will talk about this probably next time again, and then two times again, but this is a topic that, please keep bringing up, because it's definitely important. I don't... I don't... I haven't yet, thought about it much. I'm... I'm...

257

00:52:01.230 --> 00:52:26.140

Krzysztof Paruch: busy with other things in the meantime. So, yeah, so... so here you basically, like, okay, information aggregation, the way how, Hanson has introduced it, and he introduces, different, approaches here. So, we have the scoring rules, information markets, and finally the market scoring rules, which are the ones that he, proposes as being the most efficient here, or the most effective.

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00:52:26.140 --> 00:52:28.220

Krzysztof Paruch: in achieving what we're looking for.

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00:52:28.270 --> 00:52:37.220

Krzysztof Paruch: And, we had this, conversation earlier this year when we said, okay, that, like, scoring rules are very good in,

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00:52:37.230 --> 00:52:48.940

Krzysztof Paruch: rewarding or eliciting opinions from individuals, but are very poor in aggregating those opinions into an aggregate market belief. Information markets are

261

00:52:48.940 --> 00:52:57.900

Krzysztof Paruch: the opposite, so they can aggregate very well, but it's not quite as efficient as scoring rules in eliciting the truthful,

262

00:52:57.990 --> 00:52:59.539

Krzysztof Paruch: Or the, the true belief.

263

00:52:59.790 --> 00:53:22.430

Krzysztof Paruch: there are some problems with participation on information markets, like, there are some market inefficiencies, but market scoring rules seem to overcome those issues, because they, in, like, a thin market case, they behave like scoring rules, and in a thick market case, they behave like information markets. So, which is nice, right?

264

00:53:22.430 --> 00:53:40.179

Krzysztof Paruch: like, to me, this is exactly what I'm looking for. So, like, if I'm able to fully comprehend this progression from scoring rules over information markets, over markets towards market scoring rules, and if hopefully it will result in exactly kind of, like, this mechanism adjusting the

265

00:53:40.180 --> 00:53:52.450

Krzysztof Paruch: the liquidity state of the system, then I believe I should be where I wanted to be in at least being able to, implement, a market mechanism that is, yeah.

266

00:53:52.500 --> 00:54:01.930

Krzysztof Paruch: achieving the, like, upholding the properties that

I'm looking for, but also allowing people to state their opinions, and that's their beliefs.

267

00:54:04.540 --> 00:54:20.650

Krzysztof Paruch: Yeah. So, this is the definition of the scoring rules. We have, yeah, different people working on that before, and, a scoring rule is basically the function well. Now, Jim Peters mentioned in the beginning, I'm, I'm, I'm, you know.

268

00:54:21.400 --> 00:54:26.739

Krzysztof Paruch: leaning on the literature, and I'm introducing it in the same fashion. So I'm saying, okay, there is

269

00:54:26.740 --> 00:54:51.340

Krzysztof Paruch: an uncertain outcome X , and we have... it's taking values in some space, and then we have a belief, a Q over Ω , and we are asking this person to report this belief, and then we are rewarding this person based on how far their subjective belief was from the realized outcome. So the scoring rule manages to kind of, like, calculate this.

270

00:54:51.950 --> 00:55:12.959

Krzysztof Paruch: And an important property is that it's incentive compatible, right? So that we have this proper scoring rules that are eliciting truthful submissions, because they are maximizing, in expectation, the reward when agents truly report what they believe.

271

00:55:12.960 --> 00:55:16.899

Krzysztof Paruch: And the distribution P is... And over some other distribution.

272

00:55:19.870 --> 00:55:24.549

Krzysztof Paruch: So this is, that's the proper, this is the proper scoring rule.

273

00:55:24.970 --> 00:55:37.940

Krzysztof Paruch: And then some examples, so the quadratic scoring rule and the logarithmic scoring rule introduced in the 50s, so, so yeah, you can use them and just see how they work out.

274

00:55:38.580 --> 00:55:56.010

Krzysztof Paruch: Here is the whole discussion about, like, why they

are not so good, like, why they have limitations. So, for example, we have, we have a few nice, like, desirable properties, but if we, want at least two of them, we would have already dictatorship.

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00:55:56.010 --> 00:56:06.730

Krzysztof Paruch: So that, so that we couldn't really, pool opinions nicely. So, what are the, nice properties? I have it...

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00:56:09.590 --> 00:56:14.410

Krzysztof Paruch: Yeah, I think... I don't know if I wrote it up here, but it's definitely in this paper.

277

00:56:14.920 --> 00:56:28.539

Krzysztof Paruch: So... so this is why they are not good enough in pulling or in aggregating opinions. Then we have the standard information markets, which are better at doing so, but they have, some,

278

00:56:28.600 --> 00:56:41.209

Krzysztof Paruch: problems. The limitations are the thin market problems, so prices are only meaningful when substantial liquidity exists, and they have the original participation problem, so once people have, like, stated their opinion, or once they have traded once.

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00:56:41.210 --> 00:56:48.679

Krzysztof Paruch: they have no real reason to, to trade anymore. So... so any risk-averse, budget-constrained, and heterogeneous,

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00:56:48.960 --> 00:56:53.369

Krzysztof Paruch: the trader wouldn't really, interact with them. So...

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00:56:53.510 --> 00:57:12.089

Krzysztof Paruch: So this is, yeah, why information markets are not good enough. I do introduce here, en passant, like, the shares as R0 de Bro securities, which is exactly what they are, right? So, like, they pay, if an outcome is realized, and pay not if they're not.

282

00:57:12.630 --> 00:57:27.539

Krzysztof Paruch: And then, finally, we have market scoring rules, where we start with the proper scoring rule, and then out of this, we derive a market scoring rule. And it is basically paying the

difference between, the...

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00:57:27.940 --> 00:57:51.410

Krzysztof Paruch: at the stated, the stated opinion, or the stated distribution, and the, by the market at time t carried probability distribution PT . So the thing is that, here in this scenario, all the traders share a common scoring rule. So, like, contrary to, like, you know, the first case where you pay everyone according to their own scoring rule.

284

00:57:51.410 --> 00:58:05.390

Krzysztof Paruch: here you say there is an, like, a scoring rule carried by the market, and it's defined, like, with this index t , and then anyone is free to come and, like, change or update the market,

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00:58:05.390 --> 00:58:14.380

Krzysztof Paruch: opinion about this event, but they would be charged the difference, so... so they would need to either pay, or they would receive

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00:58:14.380 --> 00:58:22.700

Krzysztof Paruch: This, this difference between the scores, of the respective, Distributions, or the forecasts.

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00:58:23.710 --> 00:58:42.000

Krzysztof Paruch: Yeah, and then showing that this is also incentive, compatible, and we have, here, those two nice properties, tools for reporting, and the public forecast is updated iteratively, so we have kind of, like, this pooled, this, this pooling of information.

288

00:58:42.800 --> 00:59:07.589

Krzysztof Paruch: Right, and then, picking up again the logarithmic, rule, and deriving the logarithmic market scoring rule out of it, and then this is basically the nicest thing to have, and, like, this is where I currently am. So, demonstrating those rules, and showing, actually, that you can, do this, so you can start with the scoring rule, and then you can arrive

289

00:59:07.590 --> 00:59:08.830

Krzysztof Paruch: for the cost function.

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00:59:09.360 --> 00:59:13.480

Krzysztof Paruch: This is what, what we are... what we want to show.

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00:59:16.490 --> 00:59:33.009

Krzysztof Paruch: And this is something that I'm currently reading, which is the convexuality argument that is demonstrating this transition. So, like, you can actually do it, right? And then, key properties are incentive compatibility, boundless loss, always on liquidity, so, like, it's always possible to trade.

292

00:59:33.010 --> 00:59:42.040

Krzysztof Paruch: opinion pooling, belief aggregation, and part independence. So, irrespective of how trades happen, we will always end up in the same situation, and also with the same solvency.

293

00:59:42.570 --> 00:59:55.390

Krzysztof Paruch: So this is where I am right now, and yeah, I hope you understand the motivation behind this, and it plays into, kind of, like, answering the question of, like, how to implement now into the framework that I have developed.

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00:59:55.890 --> 01:00:01.370

Krzysztof Paruch: or the implementation, the app that I have developed, how to implement those rules to make them work.

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01:00:05.130 --> 01:00:07.530

Jamsheed Shorish: Nice, thanks. I think it's,

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01:00:08.430 --> 01:00:19.020

Jamsheed Shorish: from the point of view of kind of, like, listening and trying to understand how to incorporate this, and I think there's a lot that you're covering, which I think is exactly what kind of needs to be done, quite a lot of what you've...

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01:00:19.450 --> 01:00:38.770

Jamsheed Shorish: put together, I think, is, like, just a hair's breadth away from being, like, something that matches, like, the one pieces... the three pieces coming together, that one fits to the next one. It's like, they're just a little bit off, perhaps, in the adapters that will connect one to the other, but I think that it's there. Probably the only real question I have is, from the position of being able to

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01:00:38.770 --> 01:00:56.660

Jamsheed Shorish: address the main, what was, for me at least, the main concern of market scoring rules, which is the overwhelming computational complexity of all of the different permutations and combinations of states that are necessary to be able to estimate, as he wants, a joint probability distribution.

299

01:00:56.660 --> 01:01:10.840

Jamsheed Shorish: I think that's a thing which I would be really interested to see, kind of, like, what your... what your thinking is going to be on. It doesn't have to be answered now, obviously, and we're at time, so I wouldn't want to put you on the spot, but just a thing to sort of, like, think about is, alright, that's... that's a known concern for

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01:01:10.840 --> 01:01:20.030

Jamsheed Shorish: For being able to gain, essentially kind of the precision that you want from a market scoring rule in the event that you're not at one of the polar cases, essentially.

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01:01:20.440 --> 01:01:34.280

Jamsheed Shorish: But how do you do that? And how is it, you know, do you do, as in Hans in this paper, the idea, you know, well, we're going to restrict the probability distributions that you're allowed to choose? Or do you try to find some other method of aggregation where you restrict the aggregation methodology?

302

01:01:34.320 --> 01:01:47.939

Jamsheed Shorish: do you decide to say, hey, anybody can do anything, we've got these amazing neural networks, and you know, I'll just ask Claude to give me the proper probability distribution over that outcome? You know, these are all, like, I think, different methods by which we can kind of try to refine that.

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01:01:47.940 --> 01:01:55.720

Jamsheed Shorish: And see if that actually introduces any kind of frictions in the point of your implementation system, because eventually they have to be addressed.

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01:01:55.720 --> 01:02:10.420

Jamsheed Shorish: Because otherwise you'll end up with, you know, as Hanson had pointed out, people just won't participate. And you won't see people taking the time to do it, because the cognitive overload simply defeats them before they're able to submit a bid or submit a,

you know, any kind of prediction.

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01:02:11.090 --> 01:02:35.049

Krzysztof Paruch: Right. I mean, yeah, fully, fully agree with you, and this is just kind of, like, from a system perspective, like, what we're investigating here, but then in a second step, we can look at additional incentives, because, like, there are many ways how you can enrich those markets, or, like, those mechanisms by providing additional incentives, subsidizing those markets, you know, doing different things to ensure that

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01:02:35.510 --> 01:02:39.969

Krzysztof Paruch: like, the pure mechanistic properties.

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01:02:40.330 --> 01:02:46.549

Krzysztof Paruch: Are, are, yeah, while sufficient, but not, not kind of limiting the, the, the...

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01:02:46.700 --> 01:02:48.130

Krzysztof Paruch: Use case that we want.

309

01:02:49.450 --> 01:02:50.540

Krzysztof Paruch: Cool.

310

01:02:50.540 --> 01:02:52.020

Jamsheed Shorish: Oh, go ahead.

311

01:02:52.410 --> 01:03:02.209

Krzysztof Paruch: I need to close, because I have another meeting, so I unfortunately need to run, but it was a pleasure talking to you, and thank you so much for your valuable feedback today, and yeah, hopefully we can continue.

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01:03:02.600 --> 01:03:05.749

Jamsheed Shorish: Thanks, Chris. Yeah, I'm looking forward to the next discussion in January.

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01:03:05.750 --> 01:03:06.190

Zargham: Cheers!

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01:03:06.190 --> 01:03:06.800

Krzysztof Paruch: Bye-bye.

315

01:03:06.800 --> 01:03:07.279

Zargham: Thank you.

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01:03:07.280 --> 01:03:08.019

Jamsheed Shorish: Great, thanks everybody.

317

01:03:08.700 --> 01:03:33.660

Zargham: Actually, I sent you a few more notes in Slack, because I think there's a... it's distinct, but it's a kind of fork-off of this conversation that is back to, like, the current state of the art vis-a-vis market structure and signal processing. I think that's something we should do a session on. We just should zoom out away from the specifics of a mechanism design and focus on, like, Hanson and prediction markets.

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01:03:33.660 --> 01:03:55.330

Zargham: as well as a little bit of, again, I recommended this paper because it uses sigma algebra, or at least an approach that you'll recognize, I think, to, think about the other side of the problem. Like, what is the information process that the market structure is trying to structure? Because the engineer in me is just kind of like, I read stuff like Hansen, and I'm kind of like.

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01:03:55.330 --> 01:04:20.310

Zargham: no free lunch problem abound. It's like you're trying to, again, to prescribe a kind of... and I know that he's not, but a kind of universal mechanism, and that's the way that a lot of people who are taking up prediction markets treat it, rather than seeing it more as an innovation in the understanding of markets, rather than the mechanism, literally, that he's prescribed. And so I think there's something to be said for that criteria or requirement that we want to

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01:04:20.310 --> 01:04:45.300

Zargham: see if there's an asymmetric information process, and we are surfacing new information vis-a-vis a signal processing interpretation, and then there are actors who have structurally more access to information, and they have successfully delivered it and distilled it, then they should profit. But the other side of the problem is the actual topology and, like, the network topology of the actors and the sigma algebra of their observations.

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01:04:45.300 --> 01:05:10.269

Zargham: that's what actually characterizes the problem space that you are then applying the mechanism to try to distill information from. And so, anyway, long story short, I'm proposing that we take this topic back in one of these sessions, but that we would want to talk about, like, for example, Hansen with a focus on the requirement for the financial benefit to the parties who surface information, and then something like the

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01:05:10.270 --> 01:05:21.060

Zargham: by a paper for, like, structuring a description of that... the... of the underlying information topology. Like, who's connected to whom, and where is the information actually arriving?

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01:05:21.860 --> 01:05:28.169

Jamsheed Shorish: It would be nice to know, kind of, where the... where the state of the art actually lies on this, because so much of what's happening in the last 3 or 4

324

01:05:28.170 --> 01:05:43.219

Jamsheed Shorish: years or even less, is that the information processing capacity has just gone exponentially higher with the advent of these LLMs, and so I'm interested to see if somebody has done a spin-off work on that, and yeah, it'd be nice to see, like, maybe come January, what we see for

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01:05:43.350 --> 01:05:51.150

Jamsheed Shorish: I'm thinking, actually, from people coming back on deck or rotating in, I mean, not to put Carolina on the spot, but maybe that's something she might be interested in looking at.

326

01:05:51.150 --> 01:06:16.039

Zargham: in her areas of interest. And I do think that the first step would be, like, a directed literature review. So, like, maybe you and I could provide some pointers or some guardrails so that it's in the direction that we want, and then we could ask her if she would present on, but yeah, I mean, I think we would have to triangulate between two to three papers as the base, and then maybe do it over more than one session, but I'm just reacting

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01:06:16.040 --> 01:06:38.599

Zargham: to this from, like, just if I was my PhD supervisor, like, I'm... this way of thinking about problems is more natural for me, and

part of the reason why I tend to get frustrated with the, again, econ side of this equation is because it doesn't feel novel from the engineering standpoint. I know that it is novel in important ways, but because the engineering perspective is a bit more...

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01:06:38.600 --> 01:06:46.640

Zargham: it's just different. So, like, sometimes you're being presented ideas, and you're just kind of like, yeah, that's... you know in,

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01:06:46.730 --> 01:06:51.619

Zargham: In patents and engineering, like, the guidance that they give is, like.

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01:06:51.620 --> 01:07:16.260

Zargham: if a qualified engineer would do it, it's not novel. A lot of stuff at econ, like, trips that wire for me. And I know that that's not true, I'm just saying, like, my reaction is that, like, if a competent, like, signal processing engineer was handed the problem framed the way that it was framed, they would arrive at that, but then the problem is actually the problem framing. You're like, but actually the problem framing needs to be moved around.

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01:07:16.260 --> 01:07:17.689

Zargham: a little bit.

332

01:07:19.090 --> 01:07:37.099

Jamsheed Shorish: Yeah, I think that's a... I think that's a salient point. It would be good to know, kind of, where we are in terms of convergence of the literature strands, because I do feel that more and more we're seeing, let's say, a generational shift in economics, where they really are able to kind of interface better, or even have double majors now with economics and engineering that might be useful in leveraging.

333

01:07:37.100 --> 01:07:50.870

Zargham: The second author on that paper has two fucking PhDs, one in engineering from Penn and one in econ from MIT. That was one of my cohorts. He finished his PhD in engineering, and then he did a second fucking PhD in economics.

334

01:07:50.870 --> 01:07:51.210

Jamsheed Shorish: Excellent.

335

01:07:51.210 --> 01:08:16.059

Zargham: So, like, when I'm... I'm pointing in the same direction you are. My observation, though, is that for us as a group, I think the place where we have the most to offer is in problem framing, because of that, like, the development of a mixture of people in our group, and the fact that this knowledge is all bubbling around it. It feels like it's getting mixed together in really random ways, and just the ability to learn enough, even not individually, but collectively.

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01:08:16.060 --> 01:08:18.859

Zargham: To step back and, like, write, like.

337

01:08:18.859 --> 01:08:38.300

Zargham: I hate to say, just, like, well-posed problems for which the resulting conclusions don't need to be particularly difficult or, crazy. It's just that the missing piece is actually tying it back together and doing problem framing in a way that's suitable for, like, practical solutions.

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01:08:38.819 --> 01:08:40.259

Jamsheed Shorish: Yeah, that makes sense.

339

01:08:40.260 --> 01:08:57.229

Zargham: Anyway, I'll let you go, appreciate that. I did enjoy it, and that wasn't actually meant as a criticism, it was more of a, like, I felt, like, push to be like, oh yeah, we really, you know, I know we have too many things to think about, but, like, this feels like a place where we have, you know, unique expertise and knowledge and

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01:08:57.270 --> 01:09:08.260

Zargham: I think, just to double down on your point, you know, when Caroline is back on deck, and we're sort of scoping work for her, that's sort of like the work that she should be doing while...

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01:09:08.260 --> 01:09:19.920

Zargham: You know, when not fully booked on projects, that maybe kind of continuing to build out this thread as a research direction for her to champion is a good idea.

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01:09:19.920 --> 01:09:20.510

Jamsheed Shorish: Yeah.

343

01:09:20.620 --> 01:09:31.370

Jamsheed Shorish: Yeah, that sounds good. And of course, and since we're in the recording, I mean, kudos to Chris for continuing to flesh this out, because he's actually drawing on a tremendous amount of disparate literature to try to build that forward, so it's quite cool.

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01:09:31.370 --> 01:09:41.490

Zargham: Yeah, 100%. Yeah, exactly. That's why I say it's not a criticism, it's just that it injected some thoughts, and I'm trying to get them on the record so that we don't lose that benefit.

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01:09:41.910 --> 01:09:42.660

Jamsheed Shorish: For sure.

346

01:09:43.040 --> 01:09:43.810

Jamsheed Shorish: Cool.

347

01:09:44.020 --> 01:09:44.720

Zargham: Cheers.

348

01:09:45.020 --> 01:09:48.090

Jamsheed Shorish: Well, thanks. Thanks, and see you, Daniel. See you. See you guys.

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01:09:48.620 --> 01:09:49.390

Danilo Lessa Bernardineli: Just steal.

350

01:09:50.240 --> 01:09:50.879

Jamsheed Shorish: You bet.